NVL-TT-01 · Time Travel jeans

Men's 5-pocket straight-leg jeans with embroidery, 100 numbered pairs. Tech pack for the nominated factory.



The visual shows the placement of all details. Fit and leg shape follow the measurements on pages 3–4 and our reference jeans, which are slimmer than the visual. The wash is darker than shown (page 8).

Style	NVL-TT-01 · Time Travel · Novalife, Berlin
Quantity	100 pairs, one style, one wash, numbered 001–100 on the inner label
Sizes	W30 · W31 · W32 · W33 · W34 · W36 · W38, one inseam 81 cm (L32). Base size and proto: W32.
Fabric	12 oz rigid 100 % cotton denim, no stretch, residual shrinkage ≤ 3 % after wash
Wash	Mid-indigo vintage, honeycombs behind the knees, fading on the front thighs, no whiskers, no holes
Embroidery	4 zones on the cut panels before sewing, about 28,000 stitches (annex)
Trims	YKK metal zip, 20 mm shank button, 6 rivets, 4 snaps for the detachable back patch, all antique copper, nickel-free
Reference	Our reference jeans (size M = our W32) is sent to you after nomination. All measurements in this tech pack come from it.

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Revisions

v0.9 · 20 Oct	Quote pack (3 pages) and artwork 1:1
v1.0 draft · 21 Oct	Measurements, grading, materials, construction, process, wash, labels, QC
Next	Technical flats, belt loop positions and back pocket check from photos of the reference jeans (22–25 Oct)

This tech pack is the only valid reference. Changes only as a new version with a new date, never by chat message.

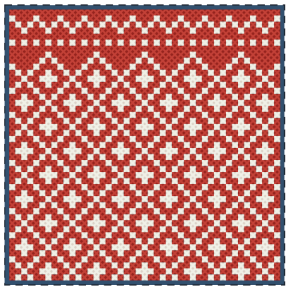
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Design overview: what goes where



A · Band, right front pocket

Cross-stitch band 23 mm wide, 17 rows, red and white. Inner edge 175 mm on the pocket opening.



E · Coin pocket

Pocket 62 × 62 mm, fully embroidered 60 × 60 mm, red and white. Centred between side seam and pocket rivet.



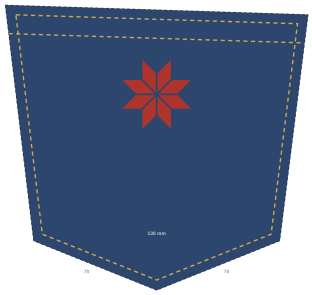
D · Back patch

Linen 80 × 71 mm, supplied by us. Detachable with 4 snaps. Back right, top flush with the waistband.



B2 · Left back pocket

Three wheat ears on a red band (48 mm). After washing: 9 gold threads 18–30 mm set by hand, knotted inside.



C · Right back pocket

Eight-point star of 8 rhombi, 36 × 36 mm, 1 mm gaps, solid red, satin stitch. Same height as the B2 band.

Hardware

Button: 20 mm shank, antique copper, eight-point star embossed.

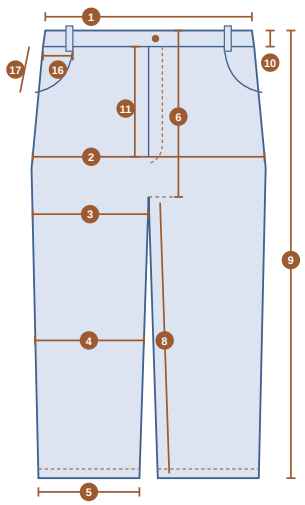
Rivets: 6 × 9 mm, plain antique copper: corners of both front pocket openings and of the coin pocket.

Snaps: 4 ring-spring snaps 10 mm for patch D (page 6).

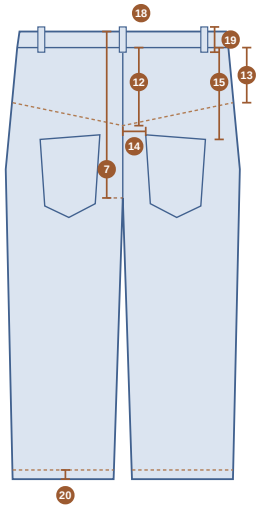
Left / right	Always as worn. The band A and coin pocket E are on the right front pocket (as worn). B2 is on the left back pocket, C and D are on the right back.
Back pockets	Top 160 mm, bottom 130 mm, sides 125 mm (inner) and 120 mm (outer), point 70 mm each side, about 150 mm high. The inner side is 5 mm higher. Same size in all sizes.
Not graded	All embroidery, coin pocket, back pockets, patch D, button, rivets and snaps are the same in every size. One embroidery program per motif.
Exact artwork	Annex, 4 pages in 1:1 scale. Print at 100 %; the check line must measure 100 mm.
Tolerance	Position of every embroidery, pocket and patch: ± 3 mm.

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Points of measure · base size W32



Front



Back

NO.	POINT OF MEASURE	HOW TO MEASURE	W32 CM	TOL.	INCH
1	**Waist, half**	Top edge of waistband, buttoned, edge to edge.	**43.0**	± 1.0	16.9
2	**Hip, half**	Straight across at the level of the end of the fly (J-stitch).	**52.0**	± 1.0	20.5
3	**Thigh, half**	One leg, 2.5 cm below crotch, at right angles to the leg.	**29.0**	± 0.6	11.4
4	**Knee, half**	One leg, 33 cm below crotch measured on the inseam, straight across.	**22.5**	± 0.5	8.9
5	**Leg opening, half**	At the hem, edge to edge.	**21.0**	± 0.5	8.3
6	**Front rise**	Crotch point to top of waistband along the seam, incl. waistband.	**32.0**	± 0.5	12.6
7	**Back rise**	Crotch point to top of waistband along the seam, incl. waistband.	**37.0**	± 0.6	14.6
8	**Inseam**	Crotch point to hem along the inseam. One length for all sizes (L32).	**81.0**	± 1.0	31.9
9	**Outseam**	Top of waistband to hem along the side seam.	**110.0**	± 1.0	43.3
10	**Waistband height**	Top to bottom edge of waistband.	**4.0**	± 0.2	1.6
11	**Fly opening**	Bottom of waistband to end of J-stitch.	**15.0**	± 0.3	5.9
12	**Back yoke at centre back**	Bottom of waistband to yoke seam, on the centre back seam.	**9.5**	± 0.3	3.7
13	**Back yoke at side seam**	Bottom of waistband to yoke seam, on the side seam.	**3.0**	± 0.3	1.2
14	**Back pocket to centre back**	Centre back seam to inner top corner of back pocket.	**3.5**	± 0.3	1.4
15	**Back pocket below waistband**	Bottom of waistband to top of back pocket, at the inner corner.	**8.5**	± 0.3	3.3
16	**Front pocket opening, width**	At the bottom of the waistband: side seam to start of pocket opening.	**11.0**	± 0.3	4.3
17	**Front pocket opening, depth**	Along the side seam: bottom of waistband to end of pocket opening.	**7.0**	± 0.3	2.8
18	**Belt loops, number**	2 front, 2 side back, 1 centre back.	**5**		
19	**Belt loop length**	Visible length, top to bottom.	**4.0**	± 0.2	1.6
20	**Hem height**	Hem edge to hem stitch line.	**1.0**	± 0.2	0.4

All measurements in cm on the **finished garment after wash**, laid flat, buttoned, not stretched. Widths 1–5 are half measurements (edge to edge, not doubled). Lengths follow the seam. Please set the shrinkage allowance of the pattern from your own wash test on our fabric.

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Grading W30–W38 (proposal)

Grade per 1" of waist size. Between W34, W36 and W38 the step is 2". Please check against your own block and send us your comments before the proto.

NO.	POINT OF MEASURE	TOL.	GRADE / 1"	W30	W31	W32	W33	W34	W36	W38
1	Waist, half	± 1.0	+1.27	40.5	41.7	**43.0**	44.3	45.5	48.1	50.6
2	Hip, half	± 1.0	+1.27	49.5	50.7	**52.0**	53.3	54.5	57.1	59.6
3	Thigh, half	± 0.6	+0.6	27.8	28.4	**29.0**	29.6	30.2	31.4	32.6
4	Knee, half	± 0.5	+0.4	21.7	22.1	**22.5**	22.9	23.3	24.1	24.9
5	Leg opening, half	± 0.5	+0.3	20.4	20.7	**21.0**	21.3	21.6	22.2	22.8
6	Front rise	± 0.5	+0.3	31.4	31.7	**32.0**	32.3	32.6	33.2	33.8
7	Back rise	± 0.6	+0.4	36.2	36.6	**37.0**	37.4	37.8	38.6	39.4
8	Inseam	± 1.0	–	81.0	81.0	**81.0**	81.0	81.0	81.0	81.0
9	Outseam	± 1.0	+0.3	109.4	109.7	**110.0**	110.3	110.6	111.2	111.8
10	Waistband height	± 0.2	–	4.0	4.0	**4.0**	4.0	4.0	4.0	4.0
11	Fly opening	± 0.3	+0.2	14.6	14.8	**15.0**	15.2	15.4	15.8	16.2
12	Back yoke at centre back	± 0.3	+0.1	9.3	9.4	**9.5**	9.6	9.7	9.9	10.1
13	Back yoke at side seam	± 0.3	–	3.0	3.0	**3.0**	3.0	3.0	3.0	3.0
14	Back pocket to centre back	± 0.3	–	3.5	3.5	**3.5**	3.5	3.5	3.5	3.5
15	Back pocket below waistband	± 0.3	–	8.5	8.5	**8.5**	8.5	8.5	8.5	8.5
16	Front pocket opening, width	± 0.3	+0.2	10.6	10.8	**11.0**	11.2	11.4	11.8	12.2
17	Front pocket opening, depth	± 0.3	–	7.0	7.0	**7.0**	7.0	7.0	7.0	7.0
18	Belt loops, number		–	5	5	**5**	5	5	5	5
19	Belt loop length	± 0.2	–	4.0	4.0	**4.0**	4.0	4.0	4.0	4.0
20	Hem height	± 0.2	–	1.0	1.0	**1.0**	1.0	1.0	1.0	1.0
	ESTIMATED QUANTITY (FINAL IN FEBRUARY)			**8**	**10**	**20**	**14**	**20**	**18**	**10**

Size names

Our size names run larger than the measured waist on purpose: relaxed straight fit, worn with a belt. Full waist in cm: W30 81.0 · W31 83.4 · W32 86.0 · W33 88.6 · W34 91.0 · W36 96.2 · W38 101.2. Size label: W.. L32.

Same in all sizes

Inseam 81 cm, waistband 4 cm, hem 1 cm, belt loops, back pocket size and position, coin pocket, all embroidery, patch D and hardware. Zip length per size: your proposal.

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Bill of materials

ITEM	DESCRIPTION	WHERE / QUANTITY	SOURCE
Shell fabric	12 oz (approx. 400 g/m²) rigid 100 % cotton denim, 3/1 twill, indigo, no stretch. Residual shrinkage ≤ 3 % in length and width after wash.	Body	Factory proposes mill and article; we approve the swatch before the proto
Pocketing	Cotton, natural white, unprinted, your standard weight.	Front pocket bags	Factory
Interlining	Waistband: your standard. Behind each snap stud: 20 × 20 mm reinforcement inside (denim or fusible).	Waistband, snaps	Factory
Topstitch thread	Corespun poly/cotton, colour copper-gold, weight your standard for jeans.	All visible topstitching and bartacks	Factory, colour card to us
Seam thread	Tonal indigo.	Inner seams, overlock	Factory
Embroidery thread	Polyester, wash- and bleach-resistant (e.g. Madeira Polyneon or equivalent). Red, white, gold (non-metallic).	A, E, B2, C	Factory, colour card to us
Embroidery backing	Cut-away backing, wash-resistant, trimmed 3–5 mm around each motif. Stitch density: your proposal in the digitizing preview.	A, E, B2, C	Factory
Gold threads B2	9 pieces, 18–30 mm, same gold as the embroidery, set by hand after wash, knotted inside the pocket.	Left back pocket	Factory
Zipper	YKK metal, antique copper finish, auto-lock slider, tonal tape. Length per size: your proposal.	Fly	Factory
Shank button	20 mm (ligne 32), antique copper, eight-point star embossed, nickel-free. Custom tooling.	Waistband, 1 pc	Factory sources, sample to us
Rivets	9 mm, plain, antique copper, nickel-free.	6 pcs: front pocket openings and coin pocket corners	Factory
Snaps	Ring-spring snaps 10 mm, antique copper, nickel-free. One system for stud, socket and cap.	4 sets: studs on jeans, sockets and caps on patch D	Factory sources and sets both parts
Patch D	Embroidered linen 80 × 71 mm, pre-shrunk, sealed edge, sew-in stiff interlining, colourfast threads, without snaps. Washable by hand on its own.	Back right	Supplied by Novalife
Inner label	Printed, permanent, readable after washing. Content on page 9.	Inside back waistband	Factory prints, text from us
Polybag	Recyclable LDPE, one pair per bag, suffocation warning in German and English.	Packing	Factory
Hangtag	Design follows (supplied by Novalife or printed by factory, to be agreed).	Left back belt loop	Open

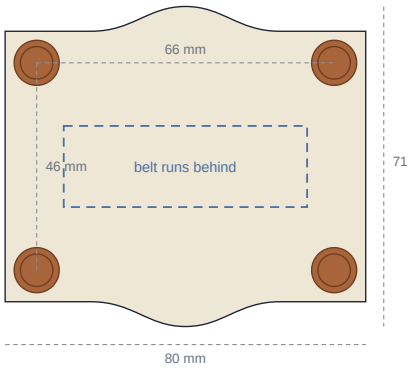
All metal parts nickel-free (EU REACH: nickel release ≤ 0.5 µg/cm²/week, EN 1811). Please send the supplier declaration or test report with the hardware samples. All trims antique copper in the same tone.

Please send for approval before the proto: fabric swatch with shrinkage result, thread colour card (topstitch and embroidery), button and snap samples. Article numbers and suppliers: please fill in and return this page.

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Construction and stitching

Stitch density	Topstitching 8–10 stitches per inch (3–4 per cm), copper-gold. Inner seams your standard.
Inseam	Flat-felled, double needle.
Side seam	Safety stitch (overlock + chain), pressed to the back, single topstitch at the top to below the front pocket.
Centre back and yoke	Flat-felled, double needle.
Front rise and fly	Zip fly, left over right as worn. J-stitch double needle, bartack at the end.
Waistband	4 cm, double needle top and bottom. Shank button 20 mm on the right end, buttonhole on the left (keyhole).
Front pockets	Double topstitch on the opening. Rivets at both ends of each opening.
Coin pocket	62 × 62 mm, embroidered panel E. Top edge double topstitch, rivets at both top corners.
Back pockets	Top hem 15 mm, double topstitch. Bartacks at both top corners (no rivets on back pockets).
Belt loops	5 loops, about 1.0–1.2 cm wide, 4 cm visible length, bartacked top and bottom. Centre back loop on the centre back seam. Positions as on the reference jeans (flats follow).
Hem	1 cm, chain stitch.
Bartacks	Copper-gold: fly end, belt loops, back pocket corners, crotch.



Patch D: detachable with 4 snaps

- Position: back right as worn, centred between the centre back loop and the right side loop. Top of the patch flush with the top of the waistband.
- Grid 66 × 46 mm between snap centres. This grid is the Novalife standard and must be exact: the patch has to fit every future Novalife jeans.
- On the jeans: 4 studs, each with a 20 × 20 mm reinforcement inside. The top two go through the waistband, the bottom two through the yoke.
- On the patch: sockets and caps, set by you on our patches.
- The belt runs between patch and jeans, between the upper and lower pair of snaps (36 mm clear). Ship every pair with the patch snapped on.
- The customer removes the patch before washing the jeans: no indigo on the linen.

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Process and approvals

NO.	STEP	WHAT	OUR APPROVAL
1	**Fabric**	Swatch and shrinkage test of the bulk fabric with our wash. You set the pattern allowance from it.	Swatch + shrinkage values
2	**Colours**	Thread colour card: topstitch copper-gold, embroidery red, white, gold.	We choose on the real thread
3	**Digitizing**	Preview of all 4 motifs: stitch count, trims, colour changes. The embroidery files (DST) belong to Novalife.	PDF/photo preview
4	**Strike-off**	Embroidery of all 4 motifs on our denim, washed with our wash recipe. No puckering or waves, red must not bleed, no frayed thread.	Photos, then physical. Gate before the proto
5	**Proto W32**	Complete jeans: embroidered, washed, all trims and patch D. Measurement sheet of points 1–20 against this tech pack.	Fit and snap test by us
6	**Wash standard**	2–3 wash trials on the proto. We choose one physical piece; it is the binding standard for bulk.	Signed wash standard
7	**PP sample**	Pre-production sample W32 in bulk fabric, bulk trims and bulk wash. Once approved it is sealed and is the standard for all 100 pairs. Measurement sheets for W30 and W38 from the first bulk cut.	Written approval before bulk
8	**TOP sample**	One finished pair from the running bulk production, sent to us physically before shipment.	Approval before shipment

Order of production

Cutting → **embroidery on the cut panels** (A front panel, E coin pocket, B2 and C back pockets) → sewing → wash → 9 gold threads by hand → rivets, button and snap studs (after wash, to protect the antique-copper finish; tell us if you recommend otherwise) → patch D snapped on → numbering → final inspection → packing.

> **Biggest risk: embroidery before the wash.** The thread must survive enzyme and bleach, and the denim shrinks under the embroidery, which can cause waves. That is why the washed strike-off (step 4) must be approved before the proto. If your test shows waves, propose pre-washing the fabric or a stabiliser.

Timeline	Proto shipped by 30 Nov 2026 · PP sample shipped by 4 Jan 2027 · bulk from mid-February · finished by 14 May 2027
Changes	No change to fabric, trims, wash or embroidery without our written approval.
Ownership	Design, artwork, pattern and embroidery files belong to Novalife. No reuse, resale or reproduction. Please send us the pattern and the DST files after production.

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Wash



Reference collage. Colour target: model photo, middle right. Thigh fading: top right, but without nicks. Honeycombs: bottom right. Bottom left: mood only, no holes or repairs.



Honeycomb target: high contrast behind the knees. On 12 oz we expect softer contrast; show us what your laser or hand-sanding can do.

Colour	Mid-indigo vintage with a slight grey cast. Darker than our first visual.
Must have	Honeycombs behind both knees. Fading on the front thighs.
Not wanted	Whiskers, holes, nicks or destroyed areas, sandblasting, potassium permanganate spray (PP spray).
Methods	Laser and/or hand-sanding, enzyme or ozone. Your proposal, without sandblasting and PP spray.
Embroidery	No abrasion over the embroidery zones (band A, coin pocket E, both back pockets): mask them during laser and sanding.
Tolerance	Bulk matches the approved physical wash standard in daylight; no streaks, no shade differences within the lot.
Tests	Residual shrinkage ≤ 3 % after a home wash at 30 °C. Rubbing fastness (ISO 105-X12): please send your result with the wash standard.

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Labels, packing, compliance

Inner label	Printed label inside the back waistband, permanent and readable after wash. Same position on every pair.
Side 1	NOVALIFE · NVL-TT-01 · Size W32 L32 · No. 001/100 Each pair has its own number 001–100. We send the number list per size before packing.
Side 2	100 % Baumwolle / 100 % Cotton Care symbols: wash 30 °C, do not bleach, do not tumble dry, iron medium (2 dots), do not dry clean „Vor dem ersten Tragen separat und auf links waschen. Indigo kann abfärben. Patch vor dem Waschen abnehmen.“ Made in [country]
Patch care	On the hangtag: „Patch separat von Hand waschen, 30 °C, nicht auswringen, flach trocknen, erst trocken wieder anknöpfen.“ (Hand wash the patch separately at 30 °C, do not wring, dry flat, snap on only when dry.)
Manufacturer info	Novalife · [postal address] · [e-mail]. We send the final text in January 2027 (EU General Product Safety Regulation). Style no. + serial number identify the product.
Fibre content	German wording is mandatory for Germany (EU Regulation 1007/2011). Please confirm the composition of the bulk fabric in writing.
Folding	Folded in thirds, front out, patch D snapped on and protected with tissue so the snaps do not mark the denim.
Polybag	One pair per recyclable bag, suffocation warning in German and English, sticker with size and number.
Cartons	Max. 15 kg. Marking: NOVALIFE · NVL-TT-01 · sizes and numbers inside · carton x of y.
Shipping papers	Commercial invoice and packing list with every shipment, including proto and PP. From Turkey: ATR movement certificate for the EU-Turkey customs union. HS code 6203 42 (men's cotton trousers), please confirm.

Chemicals: EU REACH applies. No banned azo dyes (Annex XVII entry 43), nickel release ≤ 0.5 µg/cm²/week on all metal parts (EN 1811), no PP spray, no sharp edges on metal parts. Please confirm with test reports or supplier declarations; an OEKO-TEX Standard 100 certificate for the fabric is welcome.

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Quality control

100 % check	Every pair is checked by the factory before packing: measurements 1, 2, 6, 7, 8 on two pairs per size, all other pairs visually.
Photos	Before shipment: one photo per numbered pair (front, back, inner label with number) and a short video of the final inspection. The balance is paid after we approve them.
Final inspection	AQL 2.5 major / 4.0 minor / 0 critical (ISO 2859-1, level II). We may send an inspector.

Critical (0 allowed)

- Holes, broken needle or sharp metal parts
- Wrong or duplicate serial number
- Snap opens by itself or stud comes loose
- Nickel-containing metal

Major

- Measurement outside tolerance
- Embroidery missing, off by more than 3 mm, puckered or bleeding
- Open or broken seams
- Wash not matching the standard
- Label, number or gold threads missing

Minor

- Loose threads, untrimmed ends
- Small marks that wash out
- Slightly uneven topstitching

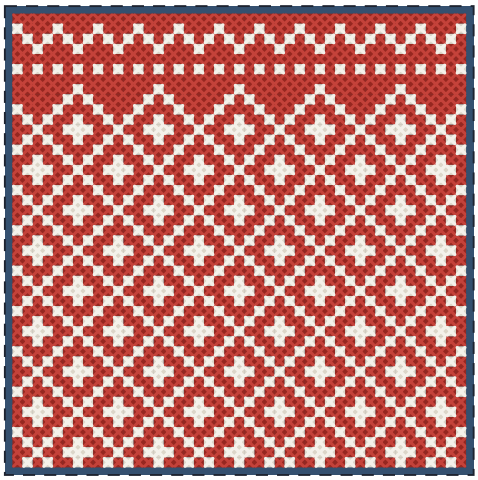
Snap and hardware test (proto and PP sample, condition for approval)

- Pull belts of 30, 35 and 40 mm width through behind patch D, 20 times each, and tighten them; sit and bend; open and close every snap 50 times. Tell us which belt widths pass easily.
- The snaps must not open by themselves and the patch must not tear. Studs, rivets and button must stay firm under a 90 N pull.
- If the 10 mm snaps fail, we switch to 12.5 mm and adjust the grid; tell us at once.
- Patch D (tested by our patch supplier): after 5 hand washes the snap grid stays 66 × 46 mm ± 0.5 mm, no fraying, no colour bleeding, interlining stays flat.

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Front · band A, coin pocket E, button, rivets

A and E belong to the wearer's right front pocket (the one with the coin pocket). The inner curved edge of A sits on the pocket opening. E covers the whole coin pocket, 62 × 62 mm: top flush with the coin pocket seam, centred between side seam and pocket rivet. Embroidery on the cut panels before sewing, polyester thread that resists the wash. Thread colours from your colour card.



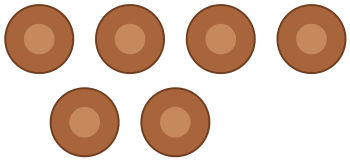
E · Coin pocket

62 × 62 mm, embroidery 60 × 60 mm
45 × 45 cross-stitches, red and white



K · Button 20 mm

shank, antique copper, embossed star



N · 6 rivets 9 mm, plain antique copper

A · Pocket band

23 mm wide, 17 cross-stitch rows, red and white
inner edge 175 mm (on the pocket opening),
outer edge 207 mm



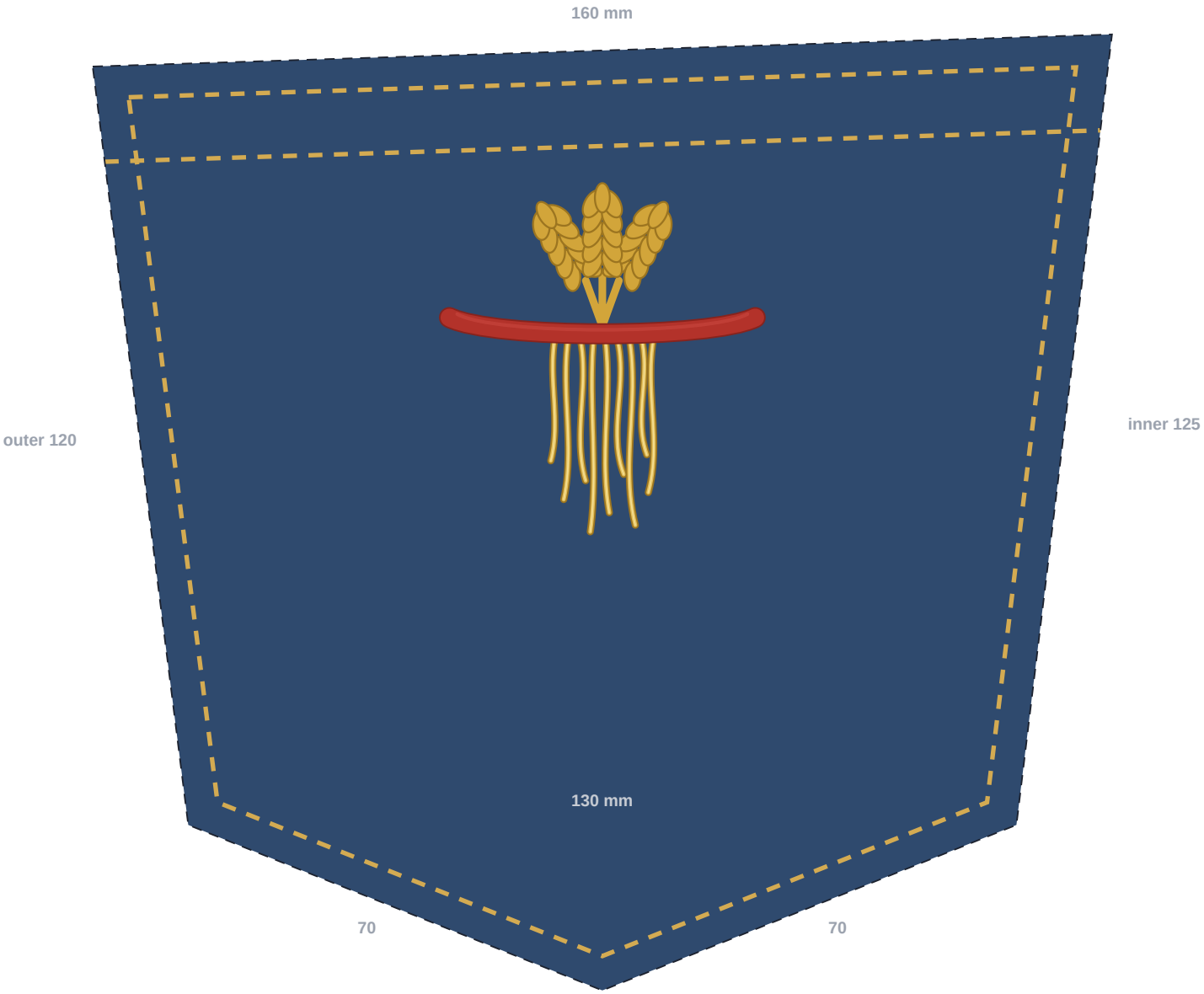
Check: this line must measure exactly 100 mm



Print at 100 % (actual size), not “fit to page”. All measurements in mm, base size M.

Back left pocket · three wheat ears on a red band (B2)

Wearer's left back pocket. The inner side (towards centre back) is 5 mm higher than the outer side.
Wheat ears and red band are embroidered. After washing: 9 gold threads, 18–30 mm, different lengths, set by hand directly under the red band and knotted inside the pocket.



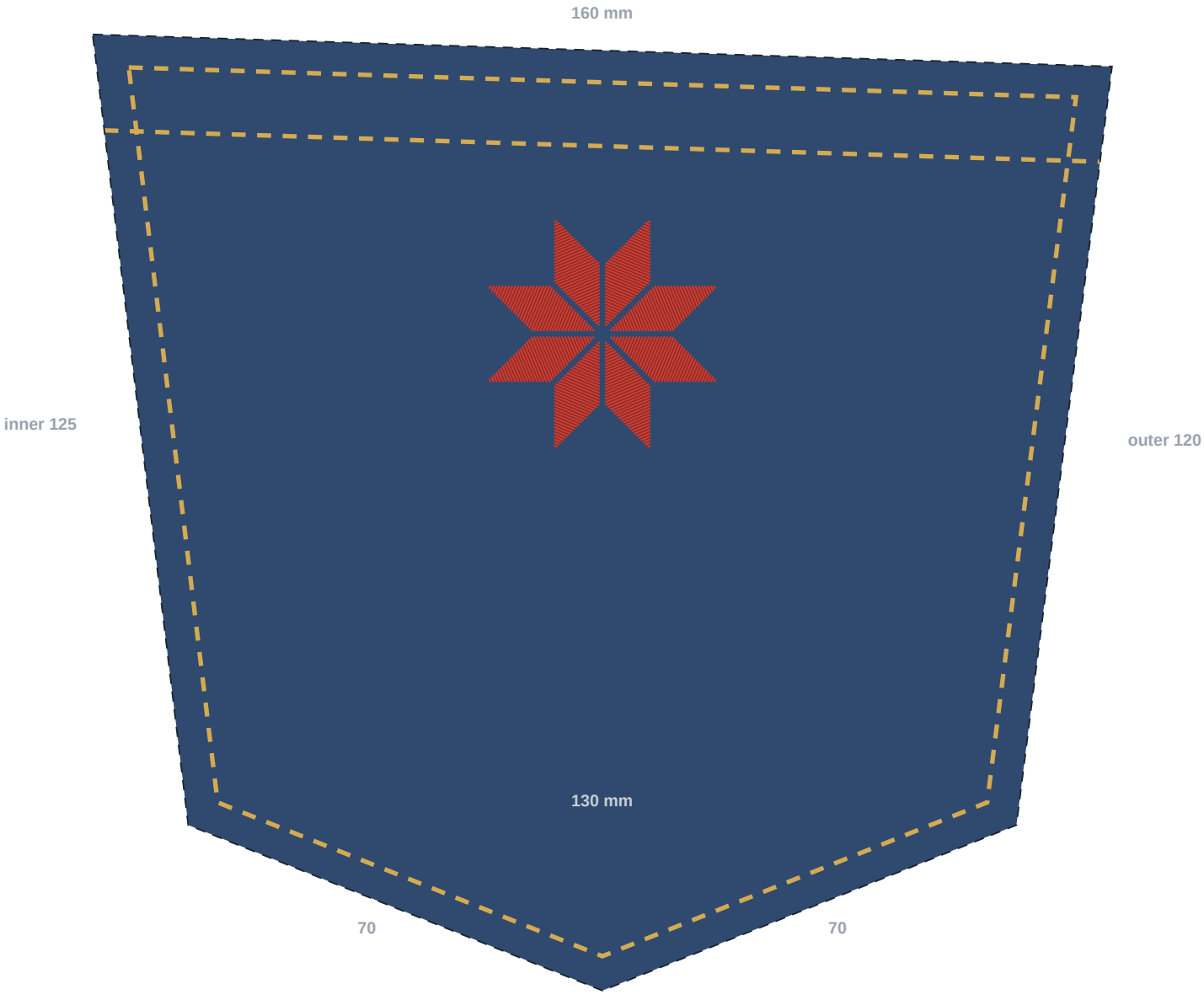
B2 · pocket top 160, bottom 130, point 70 mm each side · band 48 mm · 9 threads 18–30 mm

Check: this line must measure exactly 100 mm

Print at 100 % (actual size), not “fit to page”. All measurements in mm, base size M.

Back right pocket · eight-point star (C)

Wearer's right back pocket. The inner side (towards centre back) is 5 mm higher than the outer side.
Star centred horizontally, at the same height as the red band on the left pocket.
8 rhombi with 1 mm gaps (denim shows through), solid red, satin stitch across each rhombus.



C · pocket top 160, bottom 130, point 70 mm each side · star 36 × 36 mm, 8 rhombi, 1 mm gaps

Check: this line must measure exactly 100 mm

Print at 100 % (actual size), not “fit to page”. All measurements in mm, base size M.

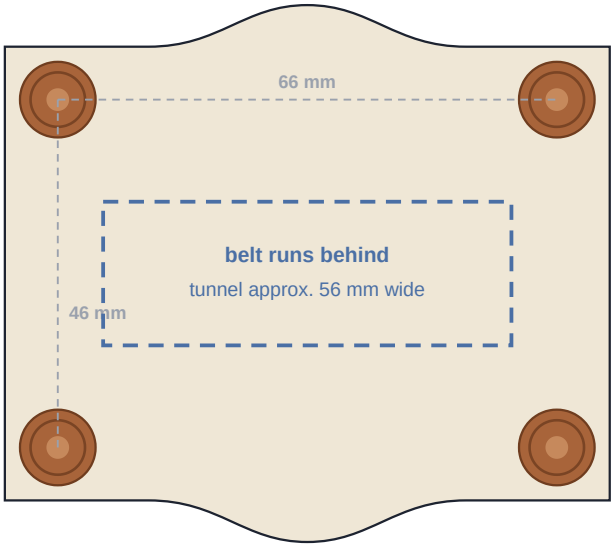
Back right · patch D, detachable with snaps

Embroidered linen patch supplied by Novalife: 80 mm wide, 71 mm high in the middle, 60 mm at the sides, max. 4 colours. Back right, centred between centre back loop and right loop. Highest point of the patch flush with the top of the waistband. Open at top and bottom, the belt runs behind the patch. Removed for washing.



D · 80 × 71 mm, sides 60 mm · artwork 1:1

Construction



4 ring-spring snaps 10 mm, antique copper, nickel-free.
Cap and socket on the patch, stud on the jeans.
Snap centres 7 mm from the side edge and the side ends.
Grid 66 × 46 mm, the same on every Novalife jeans.
Patch with stiff interlining so the snaps cannot tear out.
Jeans: reinforce each stud inside, interlining 20 × 20 mm.
Edge folded and topstitched, natural linen.
Lines min. 0.8 mm, curls min. 2.5 mm.
Approved sample from the patch maker before 100 pcs.
Shape: straight sides, 5.5 mm curve top and bottom.
Plain snaps without letters.

Check: this line must measure exactly 100 mm

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